

BEE 208 — Question 19: Skip Faulty Sensor Readings

Control Statement: for loop with continue | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables reading, total (double) and i, validCount (int)

Step 3: Set total = 0, validCount = 0

Step 4: Use a for loop from i = 1 to 12

Step 5: Prompt user to enter sensor reading i

Step 6: Read reading

Step 7: If reading == 999, display 'Faulty sensor value. Reading skipped.' and continue

Step 8: total = total + reading

Step 9: validCount = validCount + 1

Step 10: End loop

Step 11: Display validCount and average of valid readings

Step 12: End

2. PSEUDOCODE

```
BEGIN

DECLARE reading, total AS DOUBLE

DECLARE i, validCount AS INTEGER

SET total = 0, validCount = 0

FOR i = 1 TO 12 DO

    OUTPUT 'Sensor Reading', i, ':'

    INPUT reading

    IF reading == 999 THEN

        OUTPUT 'Faulty sensor value detected. Reading skipped.'

        CONTINUE

    END IF

    total = total + reading

    validCount = validCount + 1

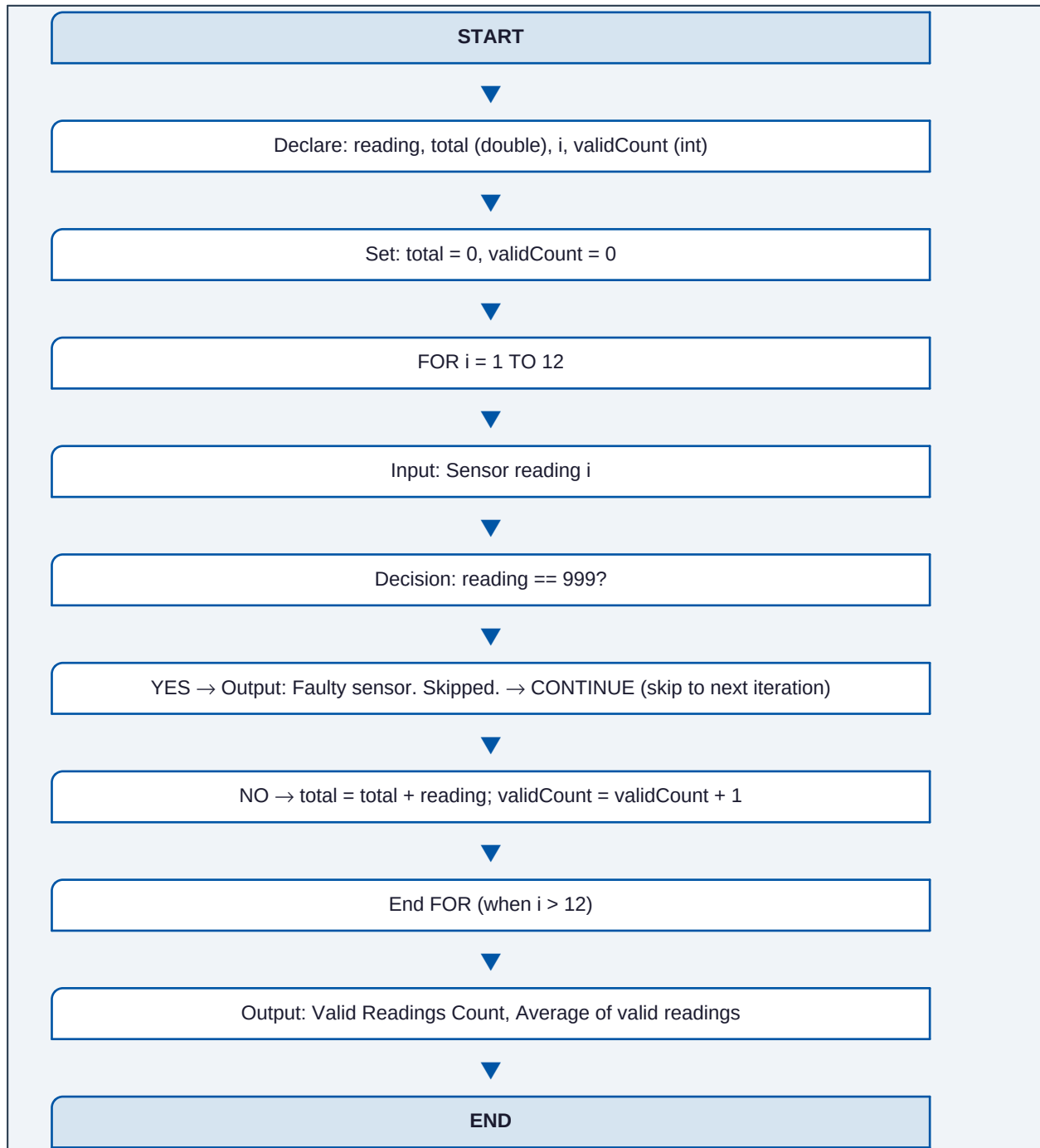
END FOR

OUTPUT 'Valid Readings:', validCount

OUTPUT 'Average:', total / validCount
```

END

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	Readings: 10,20,999,30,40,999,50,60,70,80,90,100	Expected Output	Valid=10, Average=55.0
Input	12 valid readings (no 999)	Expected Output	Valid=12, Average of all 12

